

Lesson 4 · Anti-differentiation, Rates of Change and Kinematics

NCEA Level 2 Mathematics · AS 91262 Apply calculus methods in solving problems · Handout

1 The one idea

Anti-differentiation undoes differentiation. Raise the power by 1, divide by the new power, add $+c$. Then use a known point to find c .

A derivative is a **rate of change**. In motion, velocity is the rate of change of displacement, and acceleration is the rate of change of velocity.

2 Anti-differentiating

$$x^n \longrightarrow \frac{x^{n+1}}{n+1} + c \qquad k \longrightarrow kx + c$$

$f'(x)$ term	raise the power	divide by the new power	result
$6x^2$	$6x^3$	$\frac{6x^3}{3}$	$2x^3$
$-4x$	$-4x^2$	$\frac{-4x^2}{2}$	$-2x^2$
5	$5x$	$\frac{5x}{1}$	$5x$

Why $+c$? Differentiating removes a constant, so the anti-derivative can have any constant. **Check:** differentiate your answer; you must get back $f'(x)$.

Worked example 1

Find $f(x)$ if $f'(x) = 6x^2 - 4x + 5$.

Step 1 Term by term (table above).

Step 2 $f(x) = 2x^3 - 2x^2 + 5x + c$. Check: $6x^2 - 4x + 5$.

3 Finding c

Worked example 2

$f'(x) = 4x - 3$ and the curve passes through $(2, 7)$. Find $f(x)$.

Step 1 $f(x) = 2x^2 - 3x + c$

Step 2 Substitute the point: $7 = 2(2)^2 - 3(2) + c = 2 + c$, so $c = 5$.

Step 3 $f(x) = 2x^2 - 3x + 5$

4 Rates of change

Write it in this order

1. Differentiate: the derivative is the rate.
2. Substitute the given time (if the question gives a *value of the quantity*, solve for the time first).
3. Answer with units "per ..." and the meaning: positive = increasing, negative = decreasing.

Worked example 3

The water in a tank is $V = 2000 + 30t - 0.5t^2$ litres after t minutes. Find the rate of change at $t = 10$ and at $t = 40$.

Step 1 $\frac{dV}{dt} = 30 - t$

Step 2 $t = 10$: 20, so the volume is **increasing at 20 litres per minute**.

Step 3 $t = 40$: -10 , so the volume is **decreasing at 10 litres per minute**.

Worked example 4 · given the rate, find the time

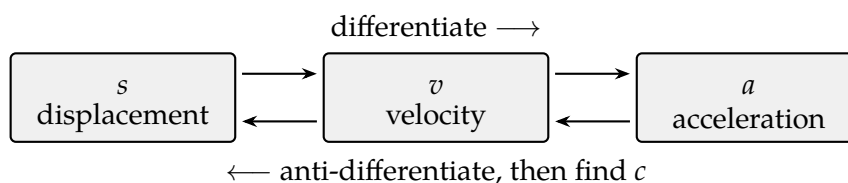
A population is $P = 500 + 40t - t^2$ after t years. When is it growing at 10 per year?

Step 1 $\frac{dP}{dt} = 40 - 2t = 10$

Step 2 $2t = 30$, so $t = 15$ years.

Sentence frame: "At $t = \dots$ the ... is increasing/decreasing at a rate of ... per ..."

5 Kinematics: displacement, velocity, acceleration



words in the question	maths
at rest, stops, changes direction	$v = 0$
maximum height	$v = 0$ (then find s)
starts from the origin / the start line	$s = 0$ when $t = 0$
starts from rest	$v = 0$ when $t = 0$
initial velocity u	$v = u$ when $t = 0$
constant acceleration -9.8 (falling)	$a = -9.8 \text{ m s}^{-2}$

Worked example 5 · differentiate

A particle moves with $s = t^3 - 6t^2 + 9t$ metres. Find (a) v at $t = 2$, (b) when it is at rest, (c) a at $t = 3$.

Step 1 $v = \frac{ds}{dt} = 3t^2 - 12t + 9$; (a) $v(2) = 12 - 24 + 9 = -3 \text{ m s}^{-1}$

Step 2 (b) $3(t - 1)(t - 3) = 0$, so **at rest at $t = 1 \text{ s}$ and $t = 3 \text{ s}$** .

Step 3 $a = \frac{dv}{dt} = 6t - 12$; (c) $a(3) = 6 \text{ m s}^{-2}$

Worked example 6 · anti-differentiate (Merit)

A ball is thrown up from 1.2 m above the ground at 14 m s^{-1} . Its acceleration is -9.8 m s^{-2} . Find its maximum height.

Step 1 $v = -9.8t + c$; $v(0) = 14$, so $v = -9.8t + 14$.

Step 2 $s = -4.9t^2 + 14t + c$; $s(0) = 1.2$, so $s = -4.9t^2 + 14t + 1.2$.

Step 3 Maximum height when $v = 0$: $t = \frac{14}{9.8} = 1.4286 \text{ s}$.

Step 4 $s = -4.9(1.4286)^2 + 14(1.4286) + 1.2 = 11.2 \text{ m}$.

6 What each grade looks like here

Achieved	anti-differentiate and find c ; find a rate at a given time; find v or a by differentiating.
Merit	find the time from a given rate or value; go from a to v to s with two constants; interpret the sign of a rate.
Excellence	a multi-step motion problem (e.g. find the cliff height from the landing speed, then the maximum height), fully explained.

7 Checking with the fx-9750GIII

In RUN-MATRIX, OPTN $\rightarrow$ CALC also has $\int dx$: the definite integral of v from 0 to 3 is the displacement over the first 3 seconds. Use it to check $s(3) - s(0)$; the working must still show $s(t)$ with its constant.

8 Ways to lose marks

From the material plus teaching observation.

- L1 Forgetting $+c$, or finding c with the wrong point (use the point the question gives).
- L2 Dividing by the old power: $x^3 \rightarrow \frac{x^4}{4}$, divide by the **new** power.
- L3 Rate of change: giving the value of the quantity instead of its derivative.
- L4 No meaning or units: say “decreasing at 528 people per hour”, not just -528 .
- L5 Kinematics: maximum height or “at rest” needs $v = 0$, not $a = 0$ or $s = 0$.
- L6 Distance is not displacement: if the object turns round, add the distance of each part.